

9 Graphics and 2D in Java

9.1 GUI in Java

When Java 1.0 was first rolled out, it only contained the abstract window toolkit (AWT), for basic GUI programming. The original AWT lacked many features that one would expect in an object-oriented toolkit:

- clipboards,
- support for printing and
- keyboard navigation.

The AWT even lacked popup menus and scroll panes – something modern GUI developer could live without. In a word, the AWT was practically useless. And you would ask yourself why? The original AWT deals with GUI elements by not dealing with them. It delegated their creation and behaviour to the native GUI toolkit on each target platform. We can say the underlying “Peer” text box actually handled the GUI. Peers are native GUI components that are manipulated by the AWT classes.

You want to create an instance of the Menu class, the Java runtime system creates an instance of the menu peer. The menu peer does the real work of displaying and managing the menu behaviour.

Under “peer” architecture, the peer – the underlying operating system takes care of painting windows and reacting to events. The AWT is like a lazy construction supervisor who attempts to look busy while he delegates all the real work to the harried construction worker bees (peers). The ease of peer architecture allowed the early developers to crank out AWT “components” that really didn’t do much. But this style of architecture means that each platform’s native window actually renders the window, and from the diagram, you can see why we call them “heavyweight” a lot was happening. And this also means slow performance. Every peer on a different operating system produced a different result. Therefore, if you used AWT on a Windows machine, your GUI looked like Windows. If you used AWT on a Macintosh, your GUI looked like Macintosh. And if you used AWT on an X11/Motif machine, your GUI crashed and burned. (X11/Motif machines could not support complex graphical elements such as menus and scrollbars.) So, using the original AWT GUI, you found that your application looked completely different on the various platforms and, what was worse, it encountered different bugs on different platforms, such as the ones on Motif. These bugs came from mismatches between the peers